

DDPG and TD3

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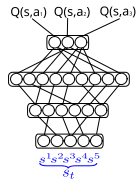
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DDPG

The Q-network in DQN

state / action	a_0	a_1	a_2	a_3
s_0	0.66	0.88*	0.81	0.73
s_1	0.73	0.63	0.9*	0.43
s_2	0.73	0.9	0.95*	0.73
s_3	0.81	0.9	1.0*	0.81
s_4	0.81	1.0*	0.81	0.9
s_5	0.9	1.0*	0.0	0.9



- ▶ Parametrized representation of the critic $\hat{Q}_{\phi}^{\pi_{\theta}}(s_t, \mathbf{a}_t)$
- ▶ Q-network equivalent to the Q-Table (with an infinity of state rows)
- ▶ For each observed $(s_t, \mathbf{a}_t, r_t, s_{t+1})$:

$$Q(s_t, \mathbf{a}_t) \leftarrow Q(s_t, \mathbf{a}_t) + \alpha [r_{t+1} + \gamma \max_{\mathbf{a} \in A} Q(s_{t+1}, \mathbf{a}) - Q(s_t, \mathbf{a}_t)]$$

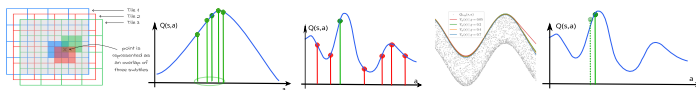
- ▶ Select action by finding $\arg \max_{\mathbf{a} \in A} Q(s, \mathbf{a})$ (as in Q-LEARNING)
- ▶ Limitation: requires one output neuron per action



Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Veness, J., Bellemare, M. G., Graves, A., Riedmiller, M., Fidjeland, A. K., Ostrovski, G., et al. (2015) Human-level control through deep reinforcement learning. *Nature*, 518(7546): 529–533.

Moving to continuous actions

- Two things become too hard:
 - Selecting actions by finding $\arg \max_{\mathbf{a} \in A} Q(\mathbf{s}, \mathbf{a})$
 - Computing $\max_{\mathbf{a} \in A} Q(\mathbf{s}_{t+1}, \mathbf{a})$ in the update rule
- That's “solving the maxQ problem”



- Five classes of solutions
 - Discretization (e.g. tile coding)
 - Use an easily optimized model (e.g. convex) (NAF, Gu et al. 2016)
 - Sample a limited set of actions (QT-Opt, Kalashnikov et al., 2018)
 - Directly approximate the max (IQL, AFU, Perrin-Gilbert, 2024)
 - DDPG: train a side estimator of the best action (also true of SAC)



Gu, S., Lillicrap, T. P., Sutskever, I., and Levine, S. (2016) Continuous deep q-learning with model-based acceleration. In Balcan, M. and Weinberger, K. Q., editors, *Proceedings of the 33rd International Conference on Machine Learning, ICML 2016, June 19-24, 2016*, volume 48 of *JMLR Workshop and Conference Proceedings*, pages 2829–2838

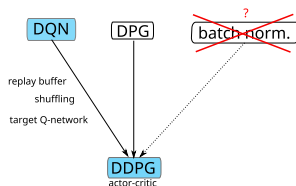


Dmitry Kalashnikov, Alex Irpan, Peter Pastor, Julian Ibarz, Alexander Herzog, Eric Jang, Deirdre Quillen, Ethan Holly, Mrinal Kalakrishnan, Vincent Vanhoucke, et al. Qt-Opt: Scalable deep reinforcement learning for vision-based robotic manipulation. *arXiv preprint arXiv:1806.10293*, 2018



Perrin-Gilbert, N. (2024) AFU: Actor-free critic updates in off-policy rl for continuous control. *arXiv preprint arXiv:2404.16159*

DDPG: ancestors



- ▶ Most of the actor-critic theory for continuous problem is for stochastic policies (policy gradient theorem, compatible features, etc.)
- ▶ DPG: an efficient gradient computation for deterministic policies, with proof of convergence
- ▶ Batch norm: inconclusive studies about impact
- ▶ Used on 32 classic control benchmarks, sometimes from pixels

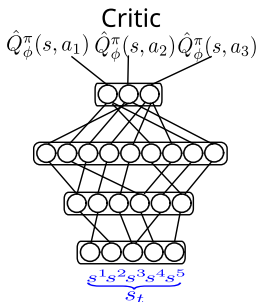


Silver, D., Lever, G., Heess, N., Degris, T., Wierstra, D., & Riedmiller, M. (2014) Deterministic policy gradient algorithms. In *ICML*

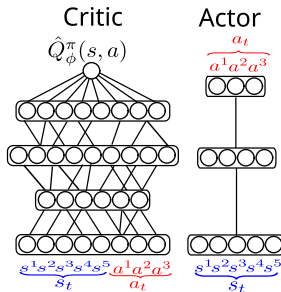


Ioffe, S. & Szegedy, C. (2015) Batch normalization: Accelerating deep network training by reducing internal covariate shift. *arXiv preprint arXiv:1502.03167*

General architecture



DQN



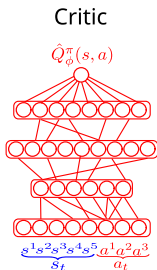
DDPG

- ▶ Actor $\pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)$, critic $\hat{Q}_{\phi}^{\pi_{\theta}}(\mathbf{s}_t, \mathbf{a}_t)$ (a single output neuron)
- ▶ All updates based on SGD
- ▶ Adaptive gradient descent techniques tune the step size (RProp, RMSProp, Adagrad, Adam...)



Lillicrap, T. P., Hunt, J. J., Pritzel, A., Heess, N., Erez, T., Tassa, Y., Silver, D., and Wierstra, D. (2015) Continuous control with deep reinforcement learning. *arXiv preprint arXiv:1509.02971* 7/9/15

Training the critic



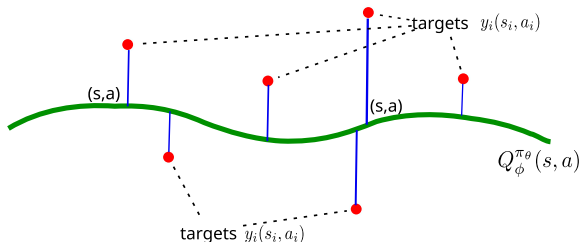
- Same idea as in DQN, but for actor-critic rather than Q-LEARNING

- Supervised learning: minimize $L(\phi) = (y^*(s, a) - \hat{F}_{\phi}(s_i, a_i))^2$
- For each sample i , the Q-network should minimize the RPE:

$$\delta_t = r_{t+1} + \gamma \hat{Q}_{\phi}^{\pi \theta}(s_{t+1}, \pi \theta(s_{t+1})) - \hat{Q}_{\phi}^{\pi \theta}(s_t, a_t)$$
- Given a minibatch of N samples $\{s_i, a_i, r_{i+1}, s_{i+1}\}$ and a target network Q' , compute $y_i = r_{i+1} + \gamma \hat{Q}'^{\pi \theta}(s_{i+1}, \pi(s_{i+1}))$
- And update ϕ by minimizing the loss function

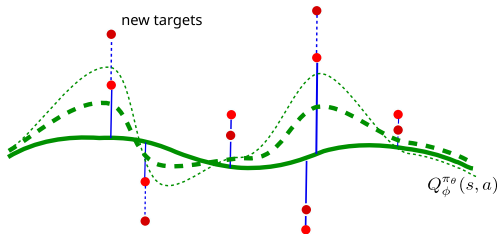
$$L = 1/N \sum_i (y_i - \hat{Q}_{\phi}^{\pi \theta}(s_i, a_i))^2$$

Learning the neural Q-function



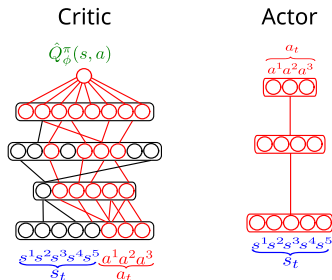
- In the tabular case, each Q-value is updated separately
- In the continuous state and action setting, interdependencies between updates
- Thus update ϕ by minimizing the squared TD loss function over minibatches

Trick 1: Stable Target Q-function



- ▶ The target $y_i = r_{i+1} + \gamma \max_a \hat{Q}_{\phi}^{\pi_{\theta}}(s_{i+1}, a)$ is itself a function of $\hat{Q}_{\phi}^{\pi_{\theta}}$
- ▶ Thus this is not truly supervised learning, and this is unstable
- ▶ Key idea: “periods of supervised learning”
- ▶ Compute the loss function from a separate *target critic* $\hat{Q}'^{\pi_{\theta}}(\dots)$
- ▶ So rather compute $y_i = r_{i+1} + \gamma \max_a \hat{Q}'^{\pi_{\theta}}(s_{i+1}, a)$
- ▶ In DQN, ϕ' is updated to ϕ only each K iterations
- ▶ In DDPG, update ϕ' using $\phi' \leftarrow (1 - \tau)\phi' + \tau\phi$ with a small gain τ

Training the actor



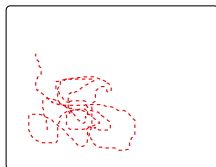
- Deterministic policy gradient theorem (Silver et al. 2014): the policy gradient is

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{s_t, a_t \sim \pi_{\theta}(\cdot)} [\nabla_a \hat{Q}_{\phi}^{\pi_{\theta}}(s_t, a_t) \nabla_{\theta} \pi_{\theta}(s_t)] \quad (1)$$

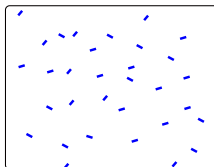
- $\nabla_a \hat{Q}_{\phi}^{\pi_{\theta}}(s_t, a_t)$ is used as error signal to update the actor weights.
- Comes from NFQCA
- $\nabla_a \hat{Q}_{\phi}^{\pi_{\theta}}(s_t, a_t)$ is a gradient **over actions**
- $y = f(w.x + b)$ (symmetric roles of weights and inputs)
- Gradient over actions $\sim$ gradient over weights



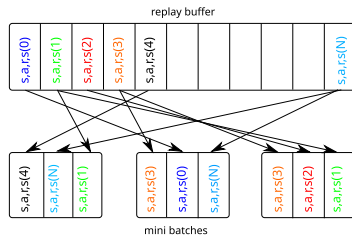
Trick2: Replay buffer shuffling



Non i.i.d. samples



i.i.d. samples

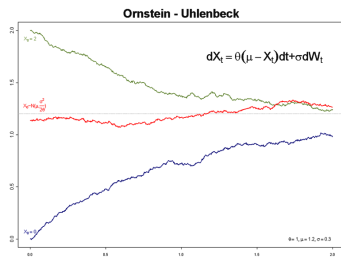


- ▶ Agent samples are not independent and identically distributed (i.i.d.)
- ▶ Shuffling a replay buffer (RB) makes them more i.i.d.
- ▶ It improves a lot the sample efficiency
- ▶ Recent data in the RB come from policies close to the current one



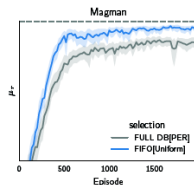
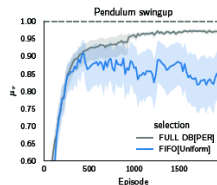
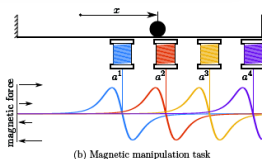
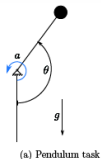
Lin, L.-J. (1992) Self-Improving Reactive Agents based on Reinforcement Learning, Planning and Teaching. *Machine Learning*, 8(3/4), 293–321

Exploration in DDPG



- ▶ The actor in DDPG is deterministic: additional exploration is needed
- ▶ The action is continuous: ϵ -greedy exploration (as in DQN) does not apply
- ▶ Solution 1: Add Gaussian noise to the actor's action
- ▶ Solution 2: Add Ornstein-Uhlenbeck (OU) noise (in the original paper)
- ▶ OU noise proved useless on the long run...

Replay buffer management



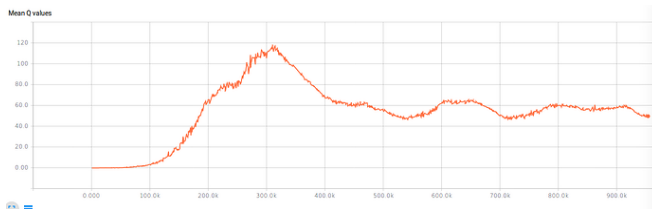
- Different replay buffer management strategies are optimal in different problems



de Bruin, T., Kober, J., Tuyls, K., & Babuška, R. (2018) Experience selection in deep reinforcement learning for control. *Journal of Machine Learning Research*, 19(9):1–56

TD3

Over-estimation with continuous actions



- ▶ All descendants of Q-learning suffer from over-estimation bias
- ▶ Even in DDPG-like algorithms which use $\hat{Q}_{\phi}^{\pi_{\theta}}(s_{t+1}, \pi_{\theta}(s_{t+1}))$ rather than a max over actions
- ▶ Clipping critic values from the knowledge of R_{max} helps, but expert knowledge required

Twin Delayed Deep Deterministic PG

- ▶ TD3 uses two critics $\hat{Q}_{\phi_1}^{\pi_\theta}$ and $\hat{Q}_{\phi_2}^{\pi_\theta}$ and target critics $\hat{Q}_{\bar{\phi}_1}^{\pi_\theta}$ and $\hat{Q}_{\bar{\phi}_2}^{\pi_\theta}$
- ▶ Compute the TD-target as the minimum to reduce over-estimation bias
- ▶ The TD-target becomes:

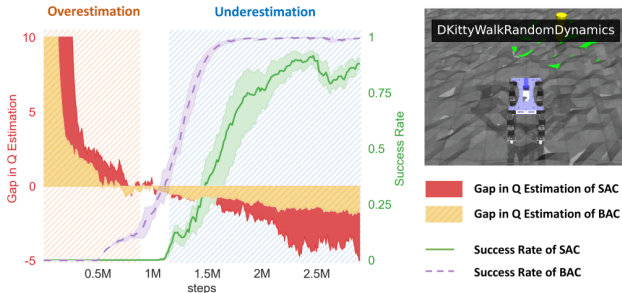
$$y_t = \mathbb{E}_{r_{t+1}, \mathbf{s}_{t+1} \sim \mathcal{D}} \left[r_{t+1} + \gamma \min_{i=1,2} \hat{Q}_{\bar{\phi}_i}^{\pi_\theta}(\mathbf{s}_{t+1}, \pi_\theta(\mathbf{s}_{t+1})) \right]$$

- ▶ To update the actor, one can use the gradient from any of the critics. E.g., $\hat{Q}_{\phi_1}^{\pi_\theta}$.
- ▶ Updating both critics at all steps is fine
- ▶ Why 2 critics and not 3 or 4? It is empirical, see TQC for more information on that...



Fujimoto, S., van Hoof, H., & Meger, D. (2018) Addressing function approximation error in actor-critic methods. *arXiv preprint arXiv:1802.09477*

Off-policy RB algorithms: remark



- ▶ DDPG, TD3 and SAC use off-policy samples to update the critic
- ▶ To update the actor, they use

$$\delta_t = \mathbf{r}_{t+1} + \gamma \hat{Q}_{\phi}^{\pi_{\theta}}(\mathbf{s}_{t+1}, \pi_{\theta}(\mathbf{s}_{t+1})) - \hat{Q}_{\phi}^{\pi_{\theta}}(\mathbf{s}_t, \mathbf{a}_t)$$
- ▶ $\hat{Q}_{\phi}^{\pi_{\theta}}(\mathbf{s}_{t+1}, \pi_{\theta}(\mathbf{s}_{t+1}))$ can be smaller than some $\hat{Q}_{\phi}^{\pi_{\theta}}(\mathbf{s}_{t+1}, a)$ present in the replay buffer
- ▶ This can give rise to underestimation



Ji, T., Luo, Y., Sun, F., Zhan, X., Zhang, J., and Xu, H. (2023) Seizing serendipity: Exploiting the value of past success in off-policy actor-critic. *arXiv preprint arXiv:2306.02865*

Any question?



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